

- 3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let $INCOME$ be household income (thousands of dollars) and $INSURANCE$ the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

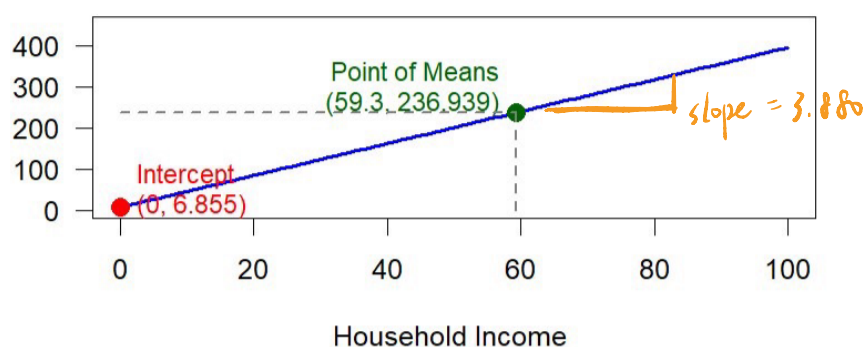
$$\widehat{INSURANCE} = 6.855 + 3.880INCOME$$

(se) (7.383) (0.112)

- Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of $INCOME = 59.3$. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.
- How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.
- Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746 .
- One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.
- Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

a. $\bar{x} = 59.3$

$$\bar{y} = 6.855 + 3.880(59.3) = 236.939$$



c. $X_0 = 100$

$$\hat{y}_0 = 6.855 + 3.880(100) = 394.855$$

$$\begin{aligned} \text{Var}(\hat{y}_0) &= \text{Var}(b_1 + b_2 X_0) \\ &= \text{Var}(b_1) + X_0^2 \text{Var}(b_2) + 2X_0 \text{Cov}(b_1, b_2) \\ &= (7.383)^2 + 100^2 (0.112)^2 + 2(100)(-0.746) \\ &= 30.7487 \end{aligned}$$

$$\text{se}(\hat{y}_0) = \sqrt{30.7487} = 5.5452$$

b. 95% C.I. ($\alpha = 0.05$)

$$b_2 \pm t_{c, df} \text{se}(b_2)$$

$$t_{0.975, 18} = 2.101$$

$$3.880 \pm 2.101(0.112) = 3.880 \pm 0.235$$

$$95\% \text{ C.I.} = [3.645, 4.115]$$

In repeated sampling, 95% of intervals would be expected to capture the true population parameter.

99% C.I. ($\alpha = 0.01$)

$$\hat{y}_0 \pm t_{c, df} \text{se}(\hat{y}_0)$$

$$t_{0.995, 18} = 2.878$$

$$394.855 \pm 2.878(5.5452) = 394.855 \pm 15.959$$

$$\therefore 99\% \text{ C.I. of expected insurance} = [378.896, 410.814]$$

The true average increase in life insurance holdings for every \$1000 increase in household income lies between \$3645 and \$4115, providing us with a 95% level of confidence in the reliability of this result.

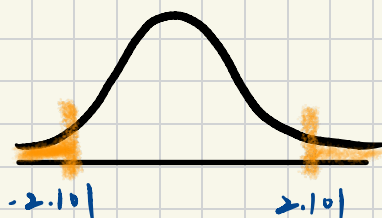
d. $H_0: \beta_2 = 5$

$H_1: \beta_2 \neq 5$

Test Statistic: $t = \frac{b_2 - \beta_2}{se(b_2)} = \frac{3.880 - 5}{0.112} = -10$

$\alpha = 5\%$

$t_{0.975, 18} = 2.101$



Since test statistic -10 falls in the rejection region, we reject H_0 .

At 5% significance level, there's sufficient evidence to suggest that $\beta_2 \neq 5$.

We deny that member's claim.

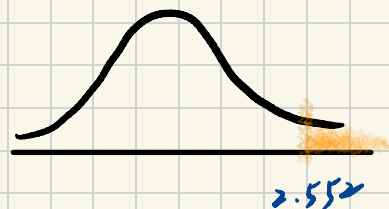
e. $H_0: \beta_2 = 1$

$H_1: \beta_2 > 1$ (right tail)

$t = \frac{b_2 - \beta_2}{se(b_2)} = \frac{3.880 - 1}{0.112} = 25.714$

$\alpha = 1\%$

$t_{0.99, 18} = 2.552$



Since test statistic 25.714 falls into the rejection region, we reject H_0 .

At 1% significance level, there's sufficient evidence to conclude that as income increases, the growth in life insurance holding exceeds the growth in income (slope β_2 is greater than 1).

HW0317

2026-03-19

#2.23

```
library(readxl)
```

```
## Warning: 套件 'readxl' 是用 R 版本 4.2.3 來建造的
```

```
collegetown <- read_excel("C:/finance ecomatrics/data/collegetown.xlsx")
View(collegetown)
```

```
collegetown$sqft2 <- collegetown$sqft^2
mod1 <- lm(price ~ sqft2, data = collegetown)
smod1 <- summary(mod1)
a1 <- coef(mod1)[[1]]
a2 <- coef(mod1)[[2]]
sea2 <- coef(smod1)[2, 2]
df <- df.residual(mod1) # 自由度 n-2
alpha <- 0.05
```

#(a): Marginal effect for a 2000 sq ft house

SQFT = 20 時的邊際效應 (ME) = $2 * \alpha_2 * 20 = 40 * \alpha_2$ $H_0: 40 * \alpha_2 \leq 13$ vs $H_1: 40 * \alpha_2 > 13$

```
ME_a <- 40 * a2
se_ME_a <- 40 * sea2
#test statistic(right tail)
t_a <- (ME_a - 13) / se_ME_a
tcr_a <- qt(1 - alpha, df)
p_value_a <- 1 - pt(t_a, df)
cat("t  =", t_a, "\ntcr =", tcr_a, "\np value =", p_value_a, "\n\n")
```

```
## t  = -26.72875
## tcr = 1.647919
## p value = 1
```

At the 5% level of significance, the calculated test statistic is $t = -26.73$, which is strictly less than the critical value $t_{cr} = 1.648$ for a right-tailed test. The p -value is 1 (greater than 0.05). Therefore, we do not reject the null hypothesis. There's sufficient evidence to conclude that for a 2000-square-foot house, the marginal effect on the expected house price of adding 100 square feet is not greater than \$13,000.

#(b): Marginal effect for a 4000 sq ft house

```
ME_b <- 80 * a2
se_ME_b <- 80 * sea2
t_b <- (ME_b - 13) / se_ME_b
tcr_b <- qt(1 - alpha, df)
p_value_b <- 1 - pt(t_b, df)
cat("t  =", t_b, "\ntcr =", tcr_b, "\np value =", p_value_b, "\n\n")
```

```
## t = 4.189467
## tcr = 1.647919
## p value = 1.65395e-05
```

At the 5% significance level, the calculated test statistic is $t = 4.189$, which strictly exceeds the critical value $t_{cr} = 1.648$, falling into the rejection region. The p -value is 1.65×10^{-5} , which is highly significant and much less than 0.05. Therefore, we reject the null hypothesis. There's sufficient evidence to conclude that for a 4000-square-foot house, the marginal effect of increasing its size by 100 square feet is significantly greater than \$13,000.

##(c): 95% Interval estimate for expected price at SQFT=20

```
new_house <- data.frame(sqft2 = 400)
ci_c <- predict(mod1, newdata = new_house, interval = "confidence", level = 0.95)
print(ci_c)
```

```
##          fit      lwr      upr
## 1 167.3735 158.0481 176.6988
```

In repeated sampling, about 95% of those intervals to contain the true population mean. Based on our sample, the interval is \$158,0481 to \$176,6988.

##(d): Sample mean of prices for houses with exactly 2000 sq ft

```
houses_2000 <- subset(collegetown, sqft == 20)
sample_mean_20 <- mean(houses_2000$price)
cat("實際樣本平均售價 =", sample_mean_20, "\n")
```

```
## 實際樣本平均售價 = 163.3333
```

The actual sample mean of selling prices for a 2000-square-foot house is calculated to be 163.33. This sample average (163.33) falls perfectly within the 95% confidence interval constructed in part (c), which is [158.05, 176.70].

#2.31

```
library(readxl)
tuna <- read_excel("C:/finance ecomatrics/data/tuna.xlsx")
View(tuna)
```

##(a): Summary Statistics & Plots

```
summary(tuna$sal1)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1762   2485   3136   6719   8612   32820
```

```
summary(tuna$apr1)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.6100  0.6900  0.8100  0.7825  0.8625  0.9200
```

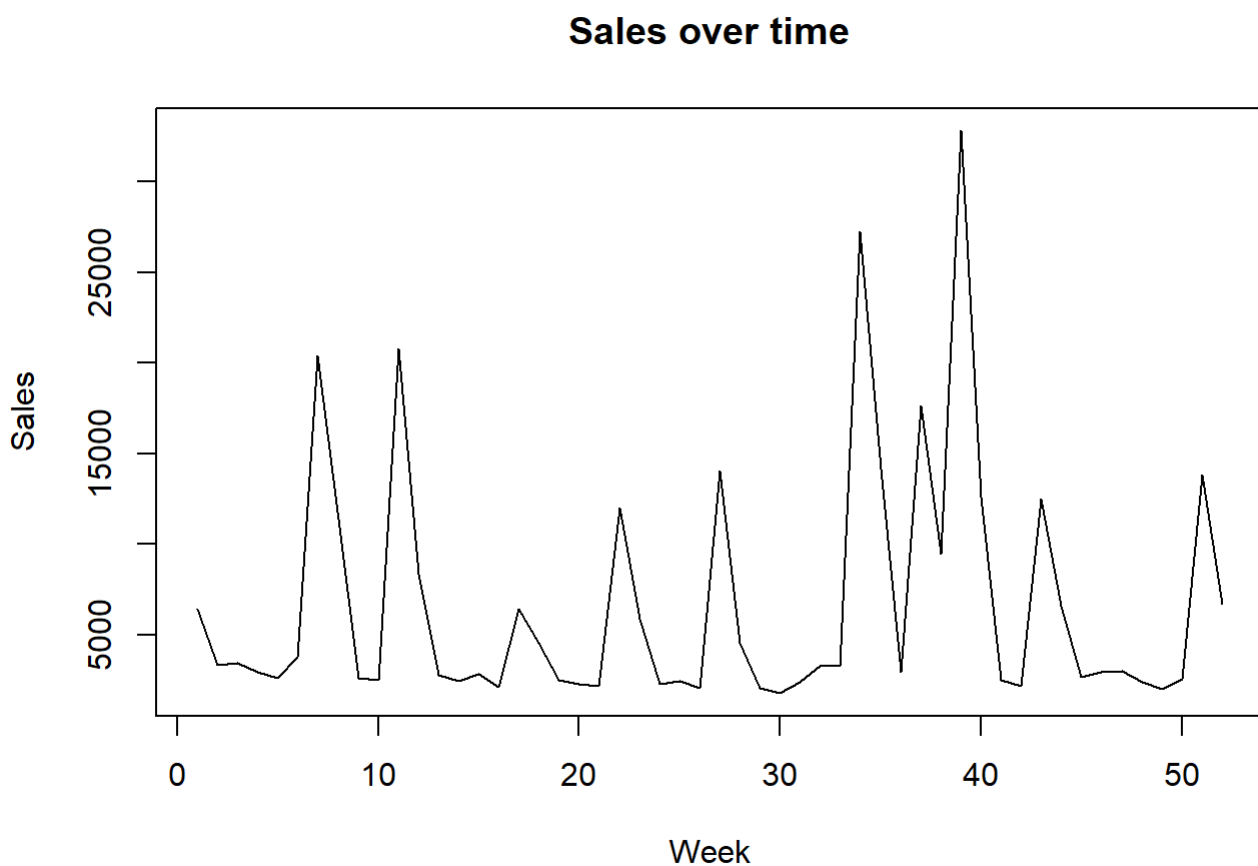
```
cat("\nstandard deviation (sal1) =", sd(tuna$sal1), "\n")
```

```
##  
## standard deviation (sal1) = 6915.083
```

```
cat("standard deviation (apr1) =", sd(tuna$apr1), "\n")
```

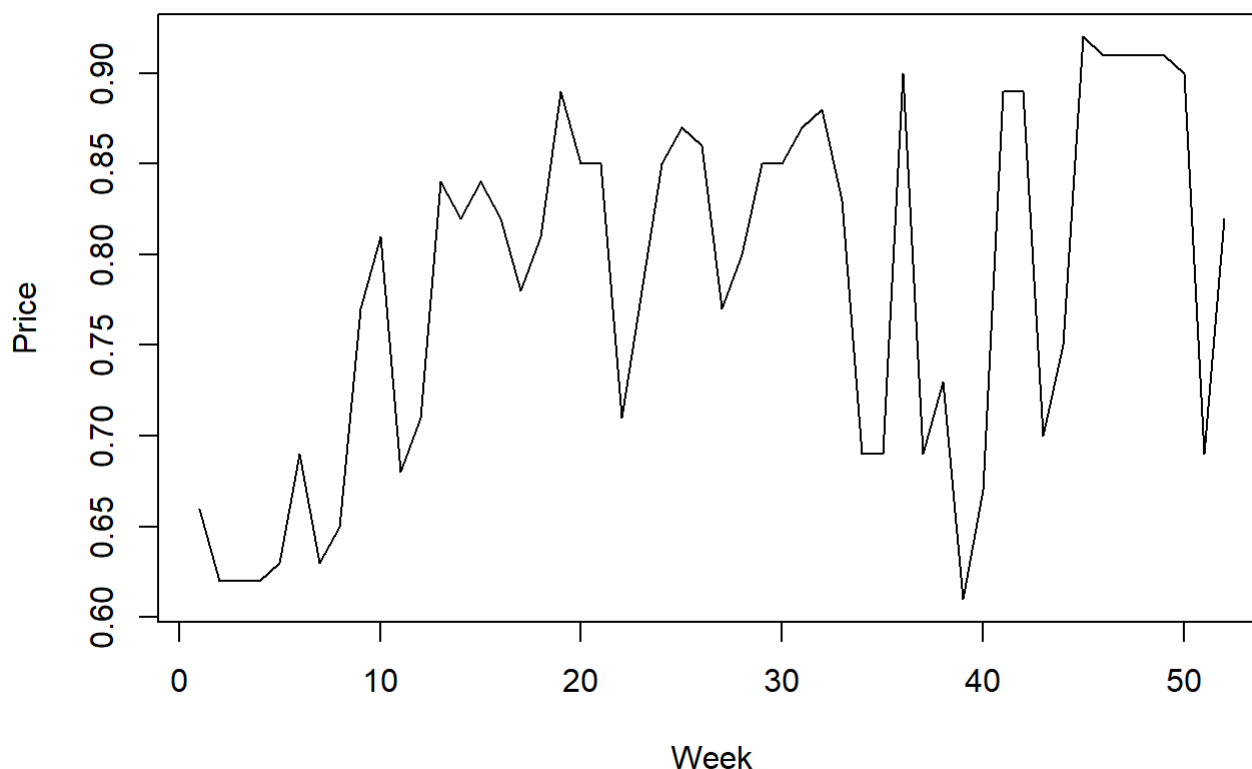
```
## standard deviation (apr1) = 0.09803711
```

```
if(!"week" %in% names(tuna)) tuna$week <- 1:nrow(tuna)  
plot(tuna$week, tuna$sal1, type="l", main="Sales over time", xlab="Week", ylab="Sales")
```



```
plot(tuna$week, tuna$apr1, type="l", main="Price over time", xlab="Week", ylab="Price")
```

Price over time



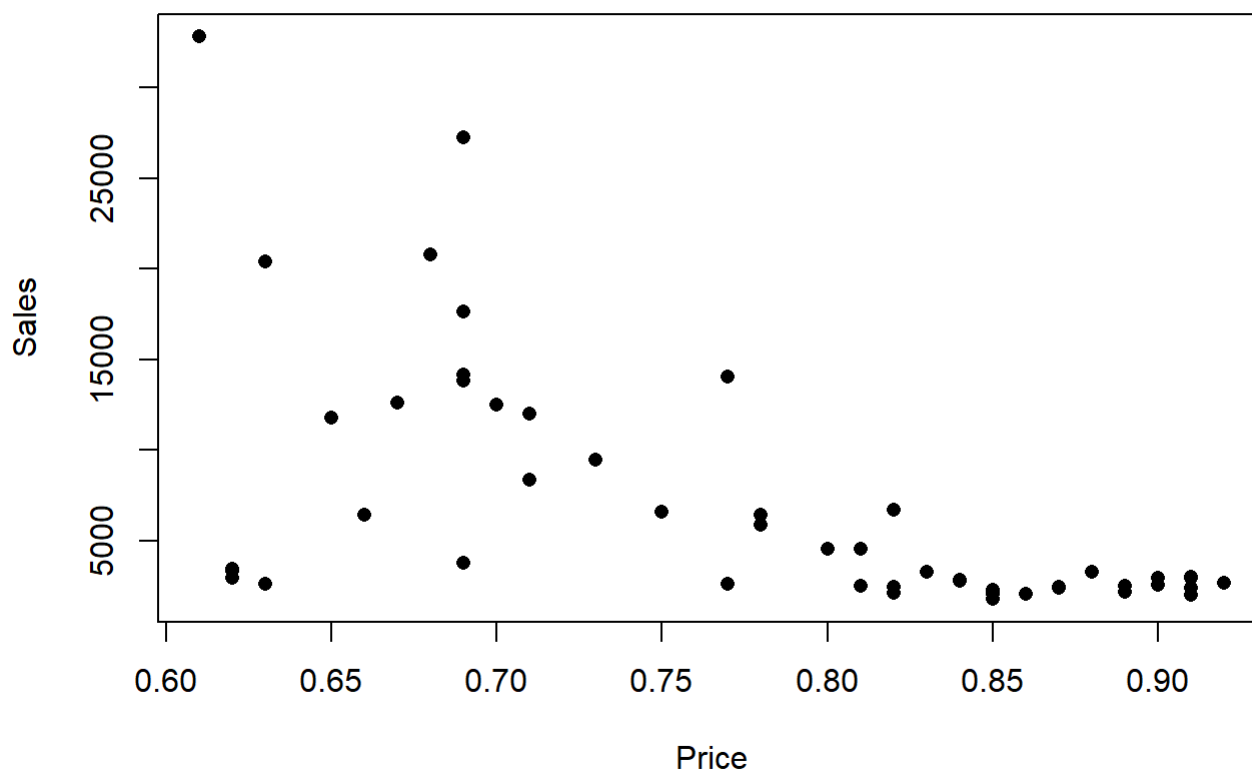
There is high week-to-week variation in both sales and price.

1. High Standard Deviation relative to the Mean: The standard deviation of sales (6,915) actually exceeds the sample mean (6,719). This indicates extreme dispersion in the weekly sales data.
2. From Plot: The time series plots are highly jagged and spiky, rather than smooth. The sales plot displays massive upward spikes (peaking over 30,000 units), indicating explosive, temporary growth in specific weeks. The price plot exhibits a pattern of frequent, sharp drops followed by quick recoveries.

#(b): Scatter Plot (SAL1 vs APR1)

```
plot(tuna$apr1, tuna$sal1, pch=16, main="Sales vs Price", xlab="Price", ylab="Sales")
```

Sales vs Price



The scatter plot clearly demonstrates a negative relationship between sales and price. This result is exactly what we would expect. According to the Law of Demand, holding all else equal, as the price of a good increases, the quantity demanded decreases, vice versa.

#(c): Point & 95% Interval Estimate for 1-cent increase

```
tuna$price1 <- 100 * tuna$apr1

mod2 <- lm(sal1 ~ price1, data=tuna)
smod2 <- summary(mod2)

b2 <- coef(mod2)[[2]]
seb2 <- coef(smod2)[2,2]

# 95% C.I. (alpha = 0.05, two-tail)
df2 <- df.residual(mod2)
tc95 <- qt(1 - 0.05/2, df2)
lowb_c <- b2 - tc95 * seb2
upb_c <- b2 + tc95 * seb2

cat("Point Estimate (b2) =", b2, "\n")
```

```
## Point Estimate (b2) = -434.4473
```

```
cat("95% CI for effect of 1-cent increase = [", lowb_c, ", ", upb_c, "]\n\n")
```

```
## 95% CI for effect of 1-cent increase = [ -592.2897 , -276.6049 ]
```

the estimated slope coefficient $\hat{\beta}_2$ is -434.45. For every 1-cent increase in the price of brand no. 1 canned tuna, the expected weekly sales volume decreases by approximately 434.45 cans.

At the 95% confidence level, the interval estimate for the true marginal effect of a 1-cent price increase on sales is [-592.29, -276.60]

#(d): 90% Interval Estimate for E(sal1 | price1=70)

```
tuna70 <- data.frame(price1 = 70)
predict(mod2, newdata = tuna70, interval = "confidence", level = 0.90)
```

```
##          fit      lwr      upr
## 1 10302.9 8624.934 11980.87
```

when the price of brand no. 1 canned tuna is 70 cents/ \$0.70, the point estimate for the expected (average) weekly sales volume is 10,302.9 can.

At the 90% confidence level, the interval estimate for the true expected number of cans sold per week at a price of 70 cents is [8,624.93, 11,980.87]

#(e) 95% Interval Estimate for Elasticity at the Means

```
mean_p <- mean(tuna$price1)
mean_s <- mean(tuna$sal1)
k <- mean_p / mean_s

E_hat <- b2 * k
se_E <- seb2 * k

tc95 <- qt(1 - 0.05/2, df2)
lowb_e <- E_hat - tc95 * se_E
upb_e <- E_hat + tc95 * se_E

cat("Point Estimate (E_hat) =", E_hat, "\n")
```

```
## Point Estimate (E_hat) = -5.059825
```

```
cat("95% CI for effect of elasticity = [", lowb_e, ", ", upb_e, "]\n\n")
```

```
## 95% CI for effect of elasticity = [ -6.898149 , -3.221501 ]
```

The point estimate for the price elasticity of demand is -5.06. This indicates that a 1% increase in the price of brand no. 1 canned tuna is expected to result in approximately a 5.06% decrease in its sales volume.

At the 95% confidence level, the interval estimate for the true population elasticity evaluated at the means is [-6.90, -3.22]

Since the absolute value of the estimated elasticity is greater than 1, we conclude that the sales are fairly elastic with respect to price.

It makes perfect economic sense. A specific brand of canned tuna has many substitutes available in the same supermarket. If Brand No. 1 raises its price, consumers incur almost zero switching costs to opt for a cheaper alternative. Consequently, the quantity demanded for this specific brand is highly sensitive to any change in its price.

#(f): Hypothesis Test for Elasticity = -3

```
# t statistic
t_stat <- (E_hat - (-3)) / se_E
# alpha=10% critical value
tcr_f <- qt(1 - 0.10/2, df2)
# p-value
p_val_f <- 2 * (1 - pt(abs(t_stat), df2))

cat("Test Statistic t =", t_stat, "\n")
```

```
## Test Statistic t = -2.250572
```

```
cat("Critical Value tcr =", tcr_f, "\n")
```

```
## Critical Value tcr = 1.675905
```

```
cat("p-value =", p_val_f, "\n")
```

```
## p-value = 0.02884204
```

```
cat("Reject H0? ->", abs(t_stat) > tcr_f, "\n")
```

```
## Reject H0? -> TRUE
```

$$H_0: E = -3 \quad H_1: E \neq -3$$

At the 10% level of significance ($\alpha = 0.10$), the critical value for a two-tailed test is $t_{cr} = 1.676$.

Rejection Region: We will reject H_0 if the calculated test statistic $t > 1.676$ or $t < -1.676$ ($|t| > 1.676$).

Since the test statistic $|-2.251|$ is greater than the critical value 1.676 (and the p-value 0.0288 is less than the significance level of 0.10), we fall into the rejection region. Therefore, we reject the null hypothesis.

At the 10% significance level, we have sufficient evidence to conclude that the elasticity of sales with respect to price is significantly different from -3.